

# Sheida Emdadi

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## Education

### M.Sc. in Computer Science

(Sep 2023 – Apr 2025)

**Department of Computing and Software, Faculty of Engineering, McMaster University,**  
Hamilton, Ontario, Canada

**Thesis:** [Communicating StateCharts \(CSC\)](#)

**Summary:** In today's increasingly networked world, almost every application depends on concurrency. However, understanding its correct implementation using commonly taught concepts is challenging, even for experienced programmers. In this thesis, we make three main contributions. First, we propose a new visual paradigm, Communicating Statecharts (CSC), that aims to simplify concurrency to the level that even novice programmers can learn and use it. Additionally, we present CSCDraw, a visual Model-Driven Development (MDD) tool to support CSC. Finally, we present the design of a pilot study to learn how best to teach CSC. This pilot works as a prelude to a rigorous study, which compares CSC to the best traditional concurrency paradigms for teaching beginners.

**Supervisors:** Dr. Spencer Smith, Dr. Christopher K. Anand

**GPA:** A+ (All courses ended with A+)

### B.Sc. in Software Engineering

(Sep 2018 – Feb 2022)

Department of Computer Engineering, Faculty of Engineering, Azad University South Tehran Branch,  
Iran

Total GPA: 3.57 /4.0 (17.37 /20)

Last 2 years GPA: 3.87 /4.0 (19.01 /20)

Last 2 semesters GPA: 3.94/4.0 (19.37/20)

## Research Interests (Pivoting)

- Software Engineering
- Reinforcement Learning
- Human-AI Interaction
- Genetic Programming
- Human Computer Interaction
- Trustworthy AI
- AI in Healthcare

## Published Papers

### Simplifying Teaching Concurrency with Communicating StateCharts (CSC)

- Journal: ACM Transactions on Computing Education (TOCE)

(Under revise for resubmission.)

## Communicating StateCharts (CSC)

Mar 2025

- Conference: ACM Innovation and Technology in Computer Science Education (ITiCSE)

## M.Sc. Academic Projects:

- **Re-implementation of GAD-EBM: Graph Anomaly Detection using Energy-Based Models.**  
Re-implemented GAD-EBM (energy-based anomaly detection on graphs) using a GCN encoder in PyTorch Geometric; built the full pipeline, edge perturbation, energy scoring via sub-graph likelihood change, and training loop, and benchmarked against 9 SOTA baselines (DOMINANT, GAAN, CONAD, etc.) on 4 real-world datasets (Weibo, Reddit, Disney, Books). Achieved ROC-AUC within ~2–9 points of the original paper's reported results while matching state-of-the-art performance among reproduced baselines.  
Code: [github.com/SheidaEmdadi/ML\\_Final](https://github.com/SheidaEmdadi/ML_Final)
  - Instructor: Dr. Lingyang Chu, Machine Learning on Graphs, Winter 2024
- **Bi-Directional Type-Inference for Algebraic Datatypes using Elm.**
  - Instructor: Dr. Christopher Anand, Advanced Type Inference, Spring 2024

## Work Experience

### Research Experience:

#### R&D Computer Scientist

June 2025– *Present*

Fondation STaBL Foundation, Nova Scotia, Canada- (Remote)

- Research and development of educational tools in the area of Computer Science.

#### Research Assistant

Jan 2025 – *Apr 2025*

Department of Computing and Software, McMaster University, Hamilton, Ontario, Canada

- Funded by MITACS..

#### Research Intern (Volunteer)

Dec 2022 – *Aug 2023*

Fondation STaBL Foundation, Nova Scotia, Canada- (Remote)

- Voluntarily Contribution on Research Projects.

### Teaching Experience:

#### Computer Science/Engineering Tutor (Freelance)

May 2020– *Present*

- Deliver instruction to undergraduate Computer Science/Software Engineering students for courses including Data Structures and Algorithms, Object-Oriented Programming, Software Design, Operating Systems, Concurrent Programming, and Databases.

#### Teaching Assistant

Sep 2024 – *Dec 2024*

Linear Optimization, Department of Computing and Software, McMaster University, Hamilton, Ontario, Canada

#### Teaching Assistant

Jan 2024 – *Apr 2024*

Computer Science Practice and Experience: Introduction to Software Design Using Web Programming, Department of Computing and Software, McMaster University, Hamilton, Ontario, Canada

Sheida Emdadi

- Lectured in review sessions and graded assignments and exams.

### Teaching Assistant

**Sep 2023 – Dec 2023**

**Introduction to Computational Thinking**, Department of Computing and Software, McMaster University, Hamilton, Ontario, Canada

- Lectured in review sessions and graded assignments and exams.

### Teaching Assistant

**Jul 2021 – Jul 2023**

**Programming Languages Design and Implementation**, Computer Engineering Department, Islamic Azad University South Tehran Branch, Tehran, Iran

- Solved students' problems with developing Android applications using Java.

### Teaching Assistant

**Jan 2021 – Dec 2021**

**Operating Systems and its Laboratory**, Computer Engineering Department, Islamic Azad University South Tehran Branch, Tehran, Iran

- Lectured in review sessions and graded assignments and exams.

### Teaching Assistant

**Jan 2021 – Dec 2021**

**Operating Systems Laboratory**, Computer Engineering Department, Islamic Azad University South Tehran Branch, Tehran, Iran

- Lectured in review sessions and graded assignments and exams.

### Teaching Assistant

**Jan 2021 – Dec 2021**

**Database Laboratory**, Computer Engineering Department, Islamic Azad University South Tehran Branch, Tehran, Iran

- Lectured in review sessions, and graded assignments.

## Computer Skills

### Programming:

- Proficient in: Java, Python, Jupyter, Elm, C, Dart (Flutter), SQL, MySQL, XML

### ML/DL Frameworks:

- PyTorch, scikit-learn, TensorFlow, PyTorch Geometric

### Data Tools:

- pandas, NumPy, matplotlib/seaborn, statsmodels

### Development Tools:

- Visual Studio Code
- Android Studio
- GitHub and GitLab

### Operating Systems:

- Linux, MS Windows

### Virtualization:

- Virtual Box

### Other:

- Firebase
- Latex
- Adobe XD (for UI, UX)
- MS Office
- Proteus

## Recent Licenses and Certificates

### **The Data Science Course:** Complete Data Science Bootcamp (2026)

(Statistical analysis, Python for ML with NumPy/pandas/scikit-learn/statsmodels, deep learning with TensorFlow, Tableau)

- Online course – Udemy

### **Supervised Machine Learning:** Regression and Classification (2025)

(Built linear and logistic regression models in Python with NumPy and scikit-learn for prediction and binary classification)

- Online course – Coursera

### **Complete Android Q Course** (2020)

(Built Android apps from scratch in Java with Firebase; covered Activities/Fragments lifecycle and UI navigation, plus SQLite and Jetpack Room for local data storage)

- Online course – Udemy

### **Flutter & Dart – The Complete Guide** (2019)

(Built cross-platform Android/iOS apps in Flutter and Dart, integrating Google Maps, device camera, authentication, and manual/automated push notifications)

- Online course – Coursera

## Language Skills

### **English:** Fluent

IELTS (General): 8 (overall score)

Listening: 7.5

Reading: 9.0

Speaking: 8.0

Writing: 8.0

Test date: June/1/2025

IELTS (Academic): 7.5 (overall score)

Listening: 8.0

Reading: 7.5

Speaking: 7.5

Writing: 6.0

Test date: July/23/2022

## Award & Honors

- First-rank graduate among about two hundred students in Systems Analysis and Design course, Fall 2020.
- First-rank graduate among about two hundred students in the Databases course, Fall 2020
- **Received an offer for a fully-funded position in Industrial Engineering at the University of Tehran, 2<sup>nd</sup> top university in Iran.**
- **Top students' award for Iranian National B.Sc. Entrance Exam, 2018.**

## M.Sc. Supervisors

**Dr. Christopher Anand**, Associate Professor, Department of Computing and Software, McMaster University

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**Dr. Spencer Smith**, Associate Professor, Department of Computing and Software, McMaster University

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